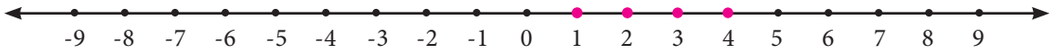
Since the solution belongs to the set of natural numbers, the solutions are 1, 2, 3 and 4. It's graph on the number line is shown below:



Example 3.14 Represent the solutions of $3x + 9 \leq 12$ in a number line, where x is an integer.

Solution

$$3x + 9 \leq 12$$

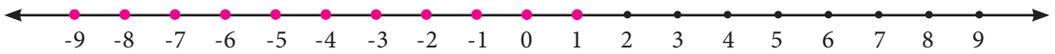
$$\frac{3x}{3} + \frac{9}{3} \leq \frac{12}{3} \quad \text{[Dividing the inequation by 3 on both sides]}$$

$$x + 3 \leq 4$$

$$x + 3 - 3 \leq 4 - 3 \quad \text{[Subtracting 3 from both sides]}$$

$$x \leq 1$$

Since the solution belongs to integers, the solutions are 1, 0, −1, −2, …. It's graph on the number line is shown below:



Example 3.15 Solve the inequation: $-2 \leq z + 3 \leq 5$, where z is an integer. Also, represent the solution, graphically.

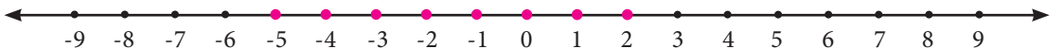
Solution

$$-2 \leq z + 3 \leq 5$$

$$-2 - 3 \leq z + 3 - 3 \leq 5 - 3 \quad \text{[Subtracting } -3 \text{ from the inequation]}$$

$$-5 \leq z \leq 2$$

Since the solution belongs to integers, the solutions are −5, −4 , −3, −2, −1, 0, 1 and 2. It's graph on the number line is shown below:



Example 3.16 Solve graphically: $6y - 5 \leq 2y + 7$, where y is an integer.

Solution

$$6y - 5 \leq 2y + 7$$

$$6y - 2y - 5 \leq 2y - 2y + 7 \quad \text{[Subtracting } 2y \text{ from both sides]}$$

$$4y - 5 \leq 7$$

$$4y - 5 + 5 \leq 7 + 5 \quad \text{[Adding 5 on both sides]}$$

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